Relational Fact Lattices for Grounding Compact Clinical Text Generators

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Abstract

A hospital pharmacy tool can explain drug interactions in plain language only when its statements are tied to dependable facts. This paper presents a Relational Fact Lattice that organizes structured relational memory for drug interaction lookup. A lightweight compact text generator writes an explanation for nurses, but first every drug pair must be matched to a lattice entry. The relations are extracted with RelForge and arranged as drug–enzyme–drug paths. Any sentence without a supporting entry is removed, producing a conservative answer. On RxInteract-5K, a pharmacist-reviewed set of 5,000 drug-pair queries, the rate of fabricated statements falls from about 21% for the generator alone to about 4% with the Relational Fact Lattice. The tool runs on a single hospital workstation CPU.

Keywords

structured relational memory, fact lattice, drug interaction lookup, confabulation reduction

1. Introduction

Drug interaction lookup is a high-consequence pharmacy task: a fluent explanation is useful only when its facts are supported. A compact text generator can provide readable wording, but unconstrained generation permits confabulation. This work introduces structured relational memory organized as a Relational Fact Lattice. It builds on RelForge [1] for extracting drug–enzyme–drug relations, then enforces a lookup-before-writing policy.

2. Method

The system stores each drug–enzyme–drug relation as a lattice entry with source sentence, confidence, and review status. RelForge extracts candidate relations and normalizes drug and enzyme names before insertion. At query time, the pharmacy interface resolves the two drug names, finds compatible paths in structured relational memory, and passes only verified entries to the compact text generator. The generator creates a short explanation with separate fields for mechanism, severity, and recommended review. A sentence filter checks every generated claim against a lattice entry; unsupported sentences are removed rather than rewritten from memory. If no path is found, the tool returns a request for pharmacist review. This policy prioritizes confabulation reduction over rhetorical completeness and keeps drug interaction lookup auditable.

3. Experiments

RxInteract-5K contains 5,000 drug-pair queries reviewed by pharmacists. The generator alone produces fabricated statements for 21.0% of queries; the Relational Fact Lattice system reduces this to 4.1%. Lookup latency is 38 ms median on a single hospital workstation CPU. The largest residual error is a missing synonym that prevents an otherwise valid pair from being found.

System	Fabricated statements	Median latency
Generator alone	21.0%	22 ms
Relational Fact Lattice	4.1%	38 ms

4. Conclusion

The Relational Fact Lattice turns structured relational memory into a practical pharmacy safeguard. RelForge supplies the relations, the compact text generator supplies readable wording, and strict support filtering achieves strong confabulation reduction. The approach is intentionally conservative and suited to drug interaction lookup on ordinary hospital hardware.

References

[1] Raman, P. (2020). RelForge: Neural Relation Extraction for Automatic Knowledge Graph Construction. Conference on Natural Language Systems (CNLS).